

1.- 30 puntos Determine si el siguiente conjunto S **forma o no una base** de $(V, +, R, *)$. Si es una base, encuentre la dimensión de V. $V = P_2(x)$.

$$S = \{\vec{v}_1 = (1 - x + x^2), \vec{v}_2 = (x - x^2)\}$$

Base

a. El conjunto debe ser linealmente independiente

$$S = \{\vec{v}_1 = (1 - x + x^2), \vec{v}_2 = (x - x^2)\}$$

$$0 + 0x + 0x^2 = \alpha(1 - x + x^2) + \beta(x - x^2)$$

$\alpha = 0$
$-\alpha + \beta = 0$
$\alpha - \beta = 0$

$\rightarrow$

1	0	0
-1	1	0
1	-1	0

$\approx$

1	0	0
-1	1	0
0	0	0

$\approx$

1	0	0
0	1	0
0	0	0

$n = r \rightarrow \text{es L.I.}$

$$\alpha = 0 \quad \beta = 0$$

es la solución trivial

el conjunto es L.I.

b) Genera al espacio al que pertenece

$$a + bx + cx^2 = \alpha(1 - x + x^2) + \beta(x - x^2)$$

$$\begin{array}{|c|} \hline \alpha = a \\ \hline -\alpha + \beta = b \\ \hline \alpha - \beta = c \\ \hline \end{array} \rightarrow \begin{array}{|c|c|c|} \hline 1 & 0 & a \\ \hline -1 & 1 & b \\ \hline 1 & -1 & c \\ \hline \end{array} \approx \begin{array}{|c|c|c|} \hline 1 & 0 & a \\ \hline 0 & 1 & a+b \\ \hline 0 & -1 & -a+c \\ \hline \end{array} \approx \begin{array}{|c|c|c|} \hline 1 & 0 & a \\ \hline 0 & 1 & a+b \\ \hline 0 & 0 & b+c \\ \hline \end{array}$$

El sistema esta condicionado a que $b+c=0$ entonces S no genera P2

No es base de P2

2.- 30 puntos Sean U y W subespacios de

- a) Encontrar una base y la dimensión de U $U = \{(a, b, c, d) \in \mathbb{R}^4 / a + b + c + d = 0\}$
- b) Encontrar la dimensión de $W = \{(a, b, c, d) \in \mathbb{R}^4 / a = -b \wedge c = 2d\}$
- c) Encontrar la dimensión $(U \cap W)$

$$U = \{(a, b, c, d) \in \mathbb{R}^4 / a + b + c + d = 0\}$$

$$W = \{(a, b, c, d) \in \mathbb{R}^4 / a = -b \wedge c = 2d\}$$

a) base y dimension U

$$a + b + c + d = 0$$

1	1	1	1
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$b, c, d = \text{libres}$

$$a = -b - c - d \quad b = \alpha \quad c = \beta \quad d = \gamma$$

$$U = \{(-\alpha - \beta - \gamma, \alpha, \beta, \gamma) \in \mathbb{R}^4 / \alpha, \beta, \gamma \in \mathbb{R}\}$$

$$(-\alpha - \beta - \gamma, \alpha, \beta, \gamma) = (-\alpha, \alpha, 0, 0) + (-\beta, 0, \beta, 0) + (-\gamma, 0, 0, \gamma)$$

$$(-\alpha - \beta - \gamma, \alpha, \beta, \gamma) = \alpha(-1, 1, 0, 0) + \beta(-1, 0, 1, 0) + \gamma(-1, 0, 0, 1)$$

$$BaseU = \{(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)\}$$

$$\dim U = n - r = 4 - 1 = 3$$

b) base y dimension de W

$a = -b$	$a + b = 0$
$c = 2d$	$c - 2d = 0$

 $\rightarrow$

1	1	0	0
0	0	1	-2

 $b, d = \text{libres}$

$$b = \alpha \quad d = \beta$$

$$W = \{(-\alpha, \alpha, 2\beta, \beta) \in R^4 / \alpha, \beta \in \mathbb{R}\}$$

$$(-\alpha, \alpha, 2\beta, \beta) = (-\alpha, \alpha, 0, 0) + (0, 0, 2\beta, \beta)$$

$$(-\alpha, \alpha, 2\beta, \beta) = \alpha(-1, 1, 0, 0) + \beta(0, 0, 2, 1)$$

$$\text{Base } W = \{(-1, 1, 0, 0), (0, 0, 2, 1)\}$$

$$\dim W = n - r = 4 - 2 = 2$$

c) base y dimension de $U \cap W$

$a + b + c + d = 0$
$a + b = 0$
$c - 2d = 0$

 $\rightarrow$

1	1	1	1	0
1	1	0	0	0
0	0	1	-2	0

 $\rightarrow$

1	1	1	1	0
0	0	-1	-1	0
0	0	1	-2	0

 $\rightarrow$

1	1	1	1	0
0	0	-1	-1	0
0	0	0	-3	0

$$\rightarrow \begin{array}{|c|c|c|c|} \hline 1 & 1 & 1 & 1 \\ \hline 0 & 0 & -1 & -1 \\ \hline 0 & 0 & 0 & -3 \\ \hline \end{array} \begin{array}{|c|} \hline 0 \\ \hline 0 \\ \hline 0 \\ \hline \end{array} \begin{array}{|c|c|c|c|} \hline 1 & 1 & 1 & 1 \\ \hline 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 1 \\ \hline \end{array} \begin{array}{|c|} \hline 0 \\ \hline 0 \\ \hline 0 \\ \hline \end{array} \quad b = \textit{libre}$$

$$\begin{array}{|c|} \hline a = -b - c - d \\ \hline c = -d \\ \hline d = 0 \\ \hline \end{array} \quad b = a$$

$$U \cap W = \{(-\alpha, \alpha, 0, 0) \in R^4 / \alpha, \beta \in \mathbb{R}\}$$

$$(-\alpha, \alpha, 0, 0) = \alpha(-1, 1, 0, 0)$$

$$\textit{Base} U \cap W = \{(-1, 1, 0, 0)\} \quad \dim U \cap W = n - r = 4 - 3 = 1$$

B) T es invertible T^{-1}

Sea $f: R^3 \rightarrow R^3$ una función vectorial definida por:

$$T \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{bmatrix} 3a \\ a - b \\ 2a + b + c \end{bmatrix} \text{ verifique si:}$$

- a) T es una transformación lineal
- b) T es invertible T^{-1}

$$T \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{bmatrix} 3a \\ a - b \\ 2a + b + c \end{bmatrix}$$

$$\text{a)} T_{(\vec{0})} = \vec{0} \quad T \left(\begin{array}{|c|} \hline 0 \\ \hline 0 \\ \hline 0 \\ \hline \end{array} \right) = \left[\begin{array}{|c|} \hline 3(0) \\ \hline 0 - 0 \\ \hline 2(0) + 0 + 0 \\ \hline \end{array} \right] = \begin{array}{|c|} \hline 0 \\ \hline 0 \\ \hline 0 \\ \hline \end{array}$$

$$\text{b)} T_{(\vec{v}_1 + \vec{v}_2)} = T_{(\vec{v}_1)} + T_{(\vec{v}_2)}$$

$$\vec{v}_1 = (a, b, c) \quad \vec{v}_2 = (a', b', c')$$

$$T \left(\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline \end{array} \right) = \left[\begin{array}{|c|} \hline 3a \\ \hline a - b \\ \hline 2a + b + c \\ \hline \end{array} \right]; T \left(\begin{array}{|c|} \hline a' \\ \hline b' \\ \hline c' \\ \hline \end{array} \right) = \left[\begin{array}{|c|} \hline 3a' \\ \hline a' - b' \\ \hline 2a' + b' + c' \\ \hline \end{array} \right]$$

$$T_{(\vec{v}_1 + \vec{v}_2)} = T \left(\begin{array}{|c|} \hline a + a' \\ \hline b + b' \\ \hline c + c' \\ \hline \end{array} \right) =$$

$$\begin{array}{|c|} \hline 3(a + a') \\ \hline a + a' - (b + b) \\ \hline 2(a + a') + b + b' + c + c' \\ \hline \end{array} \begin{array}{|c|} \hline 3a + 3a' \\ \hline a + a' - b - b' \\ \hline 2a + 2a' + b + b' + c + c' \\ \hline \end{array} = \begin{array}{|c|} \hline 3a \\ \hline a - b \\ \hline 2a + b + c \\ \hline \end{array} + \begin{array}{|c|} \hline 3a' \\ \hline a' - b' \\ \hline 2a' + b' + c' \\ \hline \end{array}$$

$$T \left(\begin{array}{|c|} \hline a + a' \\ \hline b + b' \\ \hline c + c' \\ \hline \end{array} \right) = T \left(\begin{array}{|c|} \hline a \\ \hline b \\ \hline c \\ \hline \end{array} \right) + T \left(\begin{array}{|c|} \hline a' \\ \hline b' \\ \hline c' \\ \hline \end{array} \right)$$

$$\text{c) } T \begin{pmatrix} ka \\ kb \\ kc \end{pmatrix} = kT \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$T \begin{pmatrix} ka \\ kb \\ kc \end{pmatrix} = \begin{bmatrix} 3ka \\ ka - kb \\ 2ka + kb + kc \end{bmatrix} \rightarrow \begin{bmatrix} k3a \\ k(a - b) \\ k(2a + b + c) \end{bmatrix} \rightarrow k \begin{bmatrix} 3a \\ a - b \\ 2a + b + c \end{bmatrix}$$

$$T \begin{pmatrix} ka \\ kb \\ kc \end{pmatrix} = kT \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

es una transformacion lineal

2. es invertible?

a)inyectiva

$3a = 0$
$a - b = 0$
$2a + b + c = 0$

 $\rightarrow$

3	0	0
1	-1	0
2	1	1

 $\approx$

1	0	0
1	-1	0
2	1	1

 $\approx$

1	0	0
0	-1	0
0	1	1

 $\approx$

1	0	0
0	-1	0
0	0	1

 $\approx$

1	0	0
0	1	0
0	0	1

$a = 0 \quad b = 0 \quad c = 0 \quad \text{sol. trivial}$

$\text{Base}_{\ker} = \{(0,0,0)\} \quad \dim \ker = n - r = 3 - 3 = 0 \quad \text{NULIDAD} = \vec{0}$

ES, INYECTIVA

b) sobreyectiva

APLICAMOS LA TRANSFORMACION A LA BASE USUAL

$$e = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T_{(1,0,0)} = (3(1), 1 - 0, 2(1) + 0 + 0) = (3, 1, 2)$$

$$T_{(0,1,0)} = (3(0), 0 - 1, 2(0) + 1 + 0) = (0, -1, 1)$$

$$T_{(0,0,1)} = (3(0), 0 - 0, 2(0) + 0 + 1) = (0, 0, 1)$$

$$A = \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline 0 & -1 & 1 \\ \hline 0 & 0 & 1 \\ \hline \end{array}$$

$\dim \text{Im} = 3$ es sobreyectiva

$$\det A = -3$$

$$\begin{array}{|c|c|c|c|c|c|} \hline 3 & 0 & 0 & 1 & 0 & 0 \\ \hline 1 & -1 & 0 & 0 & 1 & 0 \\ \hline 2 & 1 & 1 & 0 & 0 & 1 \\ \hline \end{array} \xrightarrow{MF1 \times 1/3} \begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 1/3 & 0 & 0 \\ \hline 1 & -1 & 0 & 0 & 1 & 0 \\ \hline 2 & 1 & 1 & 0 & 0 & 1 \\ \hline \end{array} \xrightarrow{\substack{AF2+F1 \times -1 \\ AF3+F1 \times -2}} \begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 1/3 & 0 & 0 \\ \hline 0 & -1 & 0 & -1/3 & 1 & 0 \\ \hline 0 & 1 & 1 & -2/3 & 0 & 1 \\ \hline \end{array} \xrightarrow{MF2 \times -1} \begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 1/3 & 0 & 0 \\ \hline 0 & 1 & 0 & 1/3 & -1 & 0 \\ \hline 0 & 0 & 1 & -1 & 1 & 1 \\ \hline \end{array}$$

$$\begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 1/3 & 0 & 0 \\ \hline 0 & 1 & 0 & 1/3 & -1 & 0 \\ \hline 0 & 1 & 1 & -2/3 & 0 & 1 \\ \hline \end{array} \xrightarrow{AF3+F2 \times -1} \begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 1/3 & 0 & 0 \\ \hline 0 & 1 & 0 & 1/3 & -1 & 0 \\ \hline 0 & 0 & 1 & -1 & 1 & 1 \\ \hline \end{array}$$

$$T^{-1} = \begin{array}{|c|c|c|c|} \hline 1/3 & 0 & 0 & a \\ \hline 1/3 & -1 & 0 & b \\ \hline -1 & 1 & 1 & c \\ \hline \end{array} \approx \begin{array}{|c|} \hline 1/3a \\ \hline 1/3a - b \\ \hline -a + b + c \\ \hline \end{array}$$